

Huy Nguyen

Full Stack Software Engineer

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PROFILE

I am a Full Stack Engineer specialising in complex data integrations, billing systems, and energy/EV infrastructure platforms. I bridge the gap between high-throughput C#/.NET backend architectures and fast-moving, modern user interfaces.

SKILLS

Backend .NET · C# · SQL

Frontend React · Typescript

Cloud & DevOps Azure · Bicep · GitHub Actions · Azure DevOps

EXPERIENCE

Full Stack Software Engineer — JETCharge · Oct 2021 – Jun 2026

CORE — Electric Vehicle (EV) Energy Management System

- Co-designed and led the frontend for a greenfield, first-in-Australia edge system that lets a non-specialist electrician commission a site without dealing directly with a convoluted configuration file.
 - Fits up to **5x more chargers** on the same power supply, opening capacity-constrained sites (apartments, fleet depots) that were previously uneconomic.
- Built a real-time local dashboard with WebSocket streaming and RESTful APIs.
 - Technicians and owners diagnose faults on-site instantly, replacing manual charger-by-charger and meter checks.
- Re-architected parts of the legacy backend from a monolith to a modular monolith and ported the communication layer to JET Charge's cloud platform.
 - Cut the engineering effort needed to support new message types.

Illuminate — EV Charging Asset Management Platform

- Engineered partner/contractor management letting third parties manage assets across multiple organisations — a dealer-style network.
 - Offloaded operational burden and opened training/subscription revenue. The platform backs Australia's largest EV operators, including **Woolworths, Team Global Express, RACV and SG Fleet**.

- Refactored the core database permission layer to centralise access control.
 - Eliminated scattered duplicate checks and developer-error risk across the codebase.
- Contributed to the uplift of the legacy UI to a modern React component system.
 - Sped up feature delivery and gave design freedom for a consistent product standard.

Zero-Touch Configuration (ZTC) — Remote CORE Commissioning

- Designed and led the ZTC system (custom React Node-UI & secure .NET cloud services) that pre-configures and registers charger hardware before shipping.
 - Removed on-site technician visits and cut commissioning **from hours to minutes**.
- Built a config sync engine with automatic push/pull and drift/conflict detection that recovers from transient network drops and prevents invalid device states.
 - Reduced config errors and engineer diagnosis time.
- Added a config pointer table & caching layer serving repeated reads without hitting slower, costlier blob storage.
 - Cut repeated-query response times to **under 3 seconds** and lower cloud query costs.

Billing — Apartment & Commercial Charging

- Led integration of Stripe billing into a legacy .NET platform supporting private chargers billed to individual residents and shared chargers billed per session, with automated quarterly reimbursement to the Owners Corporation.
 - Unlocked **hundreds of thousands in new recurring revenue** across commercial customers.
- Exposed billing APIs to the JET Charge Connect mobile app.
 - Residents self-manage usage and payments, lifting adoption and reducing manual admin.

IT Specialist — Energy Locals · Oct 2020 – Oct 2021

Customer Portal & Identity Verification

- Built the company's first white-labeled customer onboarding platform (Bootstrap & Laravel) with automated database provisioning.
 - Replaced manual setup and enabled self-serve, high-volume sign-ups with no per-customer engineering effort.
- Integrated Westpac and Equifax KYC verification, the company's first automated identity-verification flow.
 - Met utility-sector regulatory requirements while removing a manual review step.

Data Engineering & Systems Operations

- Designed C#/SQL ETL pipelines consolidating operational and compliance data into a single source.
 - Replaced ad-hoc manual consolidation.
- Administered high-availability VPS hosting core operational databases.

- Established the monitoring and patching protocols the team ran on.
- Automated infrastructure and dev-workstation provisioning.
 - Turned a manual, error-prone setup into a repeatable script and standardised engineer onboarding.

Graduate IT Specialist — Energy Locals · Oct 2019 – Oct 2020

AEMO Market Management System (MMS) Settlements

- Engineered C# programs and SQL procedures to reconcile AEMO national electricity market settlement data.
 - Ensured audit compliance and reconciliation accuracy across the company's market-facing financials.
- Built the ETL pipelines pulling grid pricing, demand, and network-load telemetry from AEMO MMS.
 - Established the company's reporting and analytics data foundation.

Systems & Infrastructure Automation

- Designed automated backup and disaster-recovery pipelines for critical databases.
 - Replaced manual backups and cut recovery risk.
- Automated provisioning scripts for technical workstations.
 - Made engineering setup repeatable instead of hand-configured.

EDUCATION

Computer and Network Engineering (Honours) · Mar 2016 – Mar 2020

RMIT University